KILNDECK · PRECIS

KilnDeck design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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Source	docs/02-concept.md @ 9f92bd8		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-30	Amish Chadha	Option A decided (KND-DEC-001 D10); 33 kg piece limit with five truss modules a side and 2.4 m track modules (D14); numbers from KND-CAL-001 v0.3
0.3	2026-10-29	Amish Chadha	Kit 1.9.1 upgrade; numbers refreshed from KND-CAL-001 v0.2 (storm clamps and tie-downs, KND-DEC-001 D12; fits, D13); the design is option A of KND-OPT-001
0.2	2026-10-03	Amish Chadha	TRL 2 and 3 precis; deck spans the trench on wall-top tracks (KND-DDR-001); constructable design (KND-DDR-002); first-order numbers from KND-CAL-001
0.1	2026-09-30	Amish Chadha	Initial scaffold

KilnDeck design precis

Carries kiln firemen on a railed rolling deck so they feed fuel without standing on the ash crust over the fire.

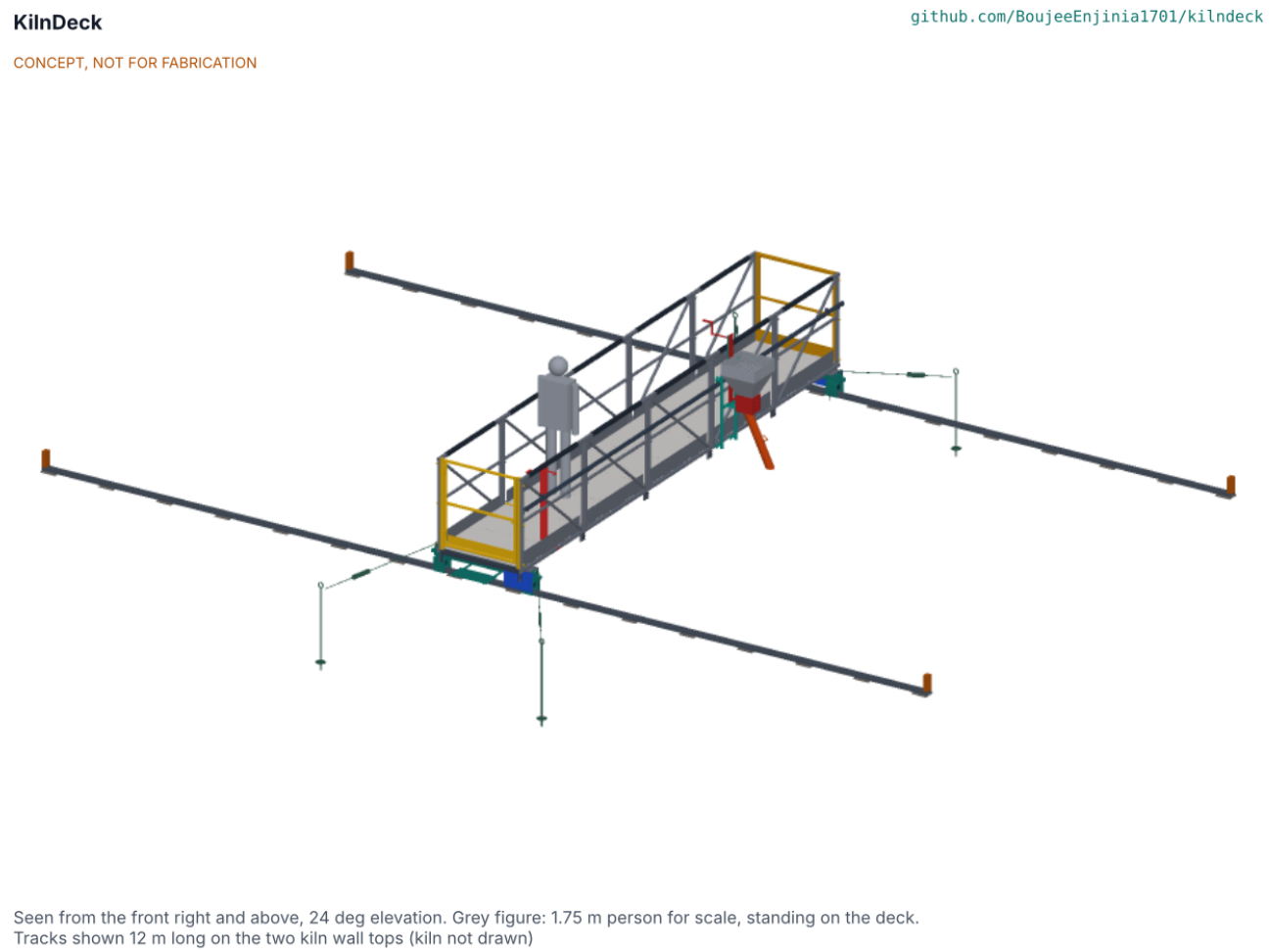


Figure 1. The deck spanning the trench on its two wall-top tracks, with the hopper carriage on the right-hand truss and a 1.75 m person on the deck for scale (concept media from the TRL 3 model).

How it works

The deck is a walkway bridge that spans the trench from the outer wall to the inner wall and rolls along the kiln on two tracks, one laid on each wall top. Nothing rests on the crust. The fireman boards from the outer wall through a self-closing gate and works inside a 1.1 m guardrail on an insulated floor about 0.4 m above the wall top. He moves the whole deck along the kiln with a hand wheel at the outer end, which drives one wheel at each end through a line shaft so both ends move together; a lock pin in the hand wheel rim holds the deck at each row of feed holes. A hopper of about 52 litres rides on a carriage along the outside of the right-hand truss, over the row of holes beside the deck. He pushes the carriage to a hole, swings the spout over it, and turns a crank above the guardrail: each quarter turn of the metering rotor drops one pocket of fuel, about 0.49 kg of coal (estimate). When the row is fed he rolls the deck about 1 m to the next row. He refills the hopper at the outer end from fuel stocked on the wall top.

Components

Table 1. Components (numbers match bom/bom.csv and the exploded view).

#	Component	Role
1	Levelling pads	Spread the track load on the wall tops, one every 0.8 m and under each joint
2	Track modules	2.4 m lengths of channel with a round bar rail, joined by sliding fishplates
3	Track end stops	Stop the deck at the ends of each track run
4	End trucks	Carry the deck ends on the tracks, with derailment guards and buffers
5	Track wheels	U-groove wheels: the outer pair guides, the inner pair floats
6	Side trusses	Span the trench; their top chords are the top rails of the guardrail
7	Cross bearers	Carry the floor and make the trusses and floor into stiff U-frames
8	Insulated floor panels	Standing surface that keeps crust heat off the feet
9	End rail and gate	Close both ends; the gate opens only inward
10	Travel drive	Hand wheel, line shaft and chain drives; the lock pin is the parking brake
11	Carriage rail	Runs along the outside of the right-hand truss
12	Hopper carriage	Carries the hopper along the deck
13	Fuel hopper	About 52 L, with a grille that keeps hands from the rotor
14	Metering rotor and crank	Drops a set dose of fuel per pocket
15	Swing spout	Directs fuel into the feed hole beside the deck
16	Grip sleeves	Heat-resistant sleeves on the top rails

Key design choices

All decided by Amish under his pre-approval of 2026-10-03 (KND-DDR-001 and KND-DDR-002).

1. **The deck spans the trench and bears only on the walls.** It reaches every hole by moving along the kiln, so no support ever stands on the crust.
2. **Movable track on the straight sides.** 2.4 m modules, about 27 kg each, are laid ahead of the fire and lifted behind it. The curved ends are not served by this prototype.
3. **The guardrail is the structure.** Two trusses 1.14 m deep carry the floor across 6.6 m, and their top chords are the top rails.
4. **Both ends driven together.** A hand wheel and line shaft stop the long deck skewing on its short wheelbase.
5. **Fuel beside the deck, not under it.** The hopper carriage feeds the next row of holes, so the floor has no openings over the fire and the fireman's feet stay at least 0.6 m from an open hole.
6. **Coal or biomass.** One rotor; the dose is set by the number of pockets.

7. **No piece over 33 kg (the HSE two-person guide; KND-DEC-001 D14, decided 2026-10-30).** Every module is carried up by two people and bolted together on the kiln, over the cold zone. Five truss modules a side and 2.4 m track modules meet it except the two end truss modules (34.4 and 34.8 kg, KND-DEC-001 D16).

First-order numbers

From KND-CAL-001 (`docs/04-calcs/sizing.py`). All are estimates for a design that has not been built.

Table 2. First-order numbers.

Quantity	Value	Assumption
Coal per fireman per round	about 30 kg	30,000 bricks a day, 1.2 MJ per kg of brick, 22 MJ/kg coal, a round every 17.5 min, two firemen
Hopper	52 L, about 42 kg of coal or 13 kg of biomass; 1.4 rounds per fill	Bulk density 0.80 and 0.25 kg/L
Dose	about 0.49 kg of coal per pocket, 1.97 kg per turn	80 % pocket fill
Deck mass	about 853 kg; heaviest piece 34.8 kg (limit 33 kg: not met by the end truss modules)	From the model solids
Bridge	11 MPa in the chords, 0.5 mm deflection	250 kg rated load at mid-span, 1.25 dynamic factor
Guardrail	91 to 100 MPa under a 1 kN push; 150 MPa at the 1.5 kN proof load	S275 steel, yield 275 MPa
Hand wheel force	68 N on level track	Rolling resistance 2 %
Wall bearing	0.154 MPa under a pad	Interim limit 0.20 MPa until the survey
Floor top temperature	34 °C at night, 48 °C in shade, 74 °C in full sun at 45 °C air	Crust 150 °C; the bare crust is the comparison
Parts cost	USD 2,905	Value-engineering target: USD 3,500. Estimated cost of the constructable design: USD 2,905 (USD 595 under the target), including USD 160 (estimate) of storm clamps and tie-downs (KND-DEC-001 D12)

Patent design-arounds

From the preliminary patent, trademark and prior-art screen (not legal advice):

- Manual metering rotor only; no motorised feeder.
- Watch any Stanford or BUET patent filing on kiln coal feeders.
- The rotating-disc charger of US1506924A (1924) has expired and is free prior art; KilnDeck uses a pocketed rotor on a horizontal shaft, not a disc.

- Watch items from the preliminary screen: rail heat and wall-top load. Both are now handled in the design (sliding track joints, floating inner wheels, pads under an interim bearing limit).

Shared blocks

- CalRig proof-load for the guardrail, deck and wall-top tracks at TRL 4.

Safety

> **Safety:** KilnDeck carries a person over a burning kiln and has moving machinery. The hazards and their controls in this design are: a fall from the deck (1.1 m top rail, knee rail and toe board on all four sides; an inward-opening self-closing gate); the deck running off its track (end stops on every track end, buffers on the trucks, derailment guards either side of each track); the deck moving while someone boards or feeds (lock pin in the hand wheel; the gate is at the hand wheel end); crushed fingers in the drive and rotor (sealed chain case, sheet chain guards, a grille over the hopper, a crank turned by hand with low force); hot surfaces (silicone grip sleeves; an insulated floor, which still reaches about 74 °C in full summer sun, so heat-resistant footwear stays necessary); and the kiln wall giving way (pads under an interim bearing limit and a structural survey of each kiln before fitting).
>
> The deck is assembled over the cold zone of the kiln and rolled to the fire only when complete. Heat exposure to the fireman remains: shade, water and rest breaks are still needed. The guardrail and deck must be proof-loaded before use (TRL 4). This design is published as an open engineering reference. It is not certified equipment.

Open questions

None for the design; all decisions are recorded in the design decisions register (KND-DEC-001). Facts that depend on the partner kiln (trench width, wall condition, feed hole grid, crust temperature) and on the parts bought are listed there to confirm.